How-To Library

Informatica

# Informatica® Big Data Management 10.2.2 on Microsoft Azure: Architecture and Best Practices

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# Abstract

You can tune Informatica® Big Data Management for better performance. This article provides sizing recommendations for a Hadoop or Databricks cluster and the Informatica domain in a cloud or hybrid deployment of Big Data Management with Microsoft Azure cloud platform. The article gives tuning recommendations for various Big Data Management and Azure components. This article is intended for Big Data Management users, such as Hadoop administrators, Informatica administrators, and Informatica developers.

# Supported Versions

- Big Data Management 10.2.2

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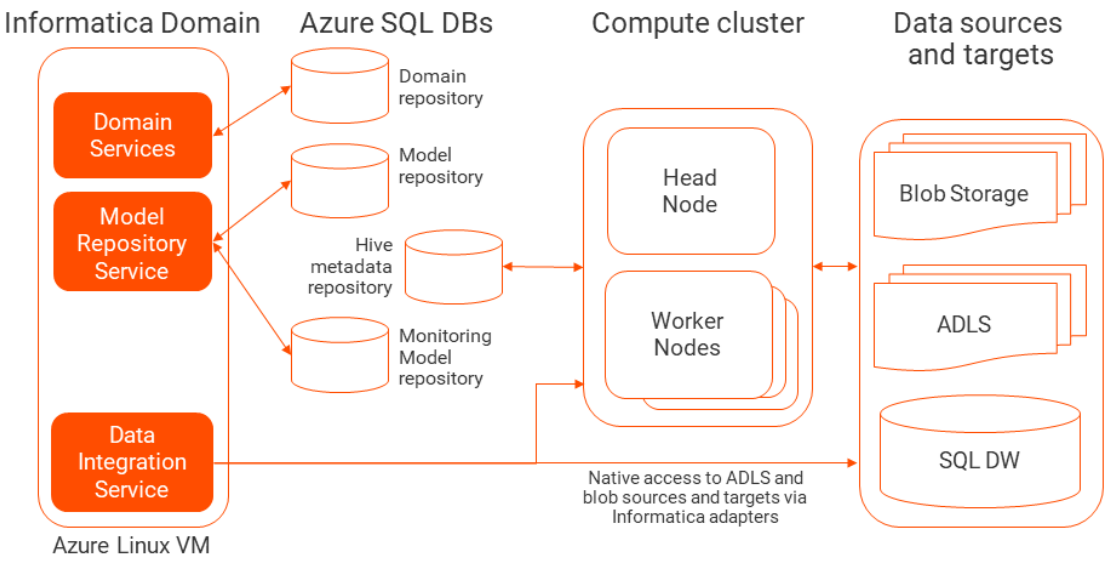
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# Overview

Customers of Microsoft Azure and Informatica can deploy Informatica Big Data Management in the Azure cloud platform to take advantage of Informatica's integration with the Azure cloud platform and Azure compute and storage resources. The compute cluster can be HDInsight or Databricks.

The following image shows the integration of Big Data Management with the Azure cloud platform and the compute cluster:




The Informatica domain may be deployed on the cloud platform or on-premises. The domain uses Azure SQL databases for the Domain repository, the Model repository, the monitoring Model repository, and the Hive metadata repository. When a mapping, application or workflow is deployed to the Data Integration Service, it can run on the compute cluster.

You can tune Big Data Management in the following areas:

- Hardware
- Compute cluster parameters
- Domain parameters and application services in the domain

- Big Data Management engines

# Big Data Management on the Azure Cloud Platform

You can deploy Big Data Management on the Azure cloud platform in the following ways:

**Manual cloud deployment**

Manually install and configure the Informatica domain and Big Data Management on Azure cloud platform VMs in the same region as your compute cluster.

**Marketplace cloud deployment**

Deploy Big Data Management from the Azure marketplace to create an Informatica domain and a compute cluster in the Azure cloud, learning about Big Data Management functionality through prepackaged mappings.

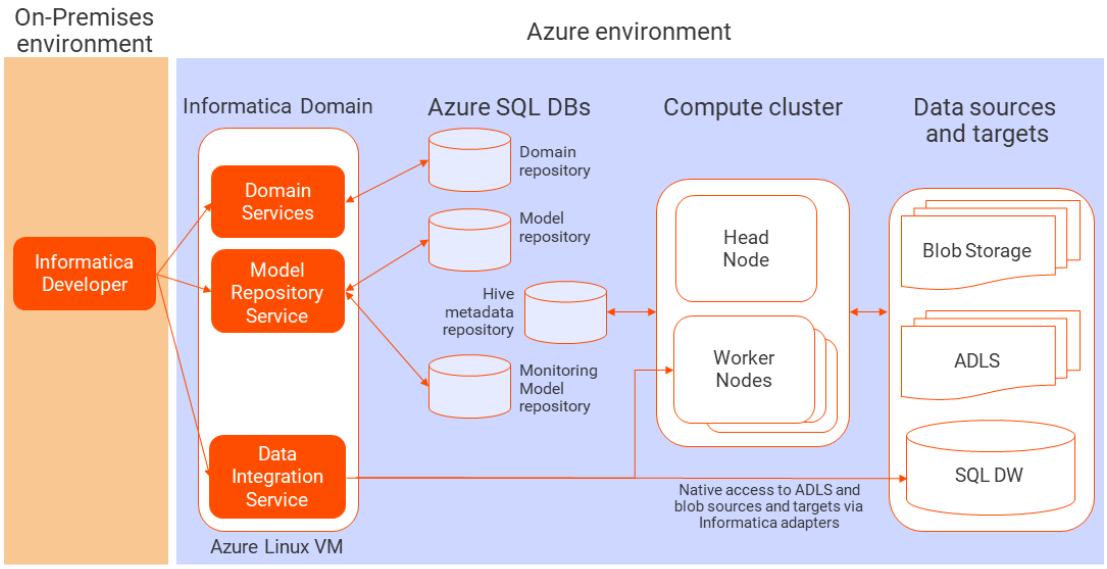
**Hybrid deployment**

Install and configure the Informatica domain and Big Data Management on-premises, and configure them to push processing to the compute cluster.

This article describes performance considerations for a manual deployment of Big Data Management on the Azure cloud platform.

## *Architecture of an Azure Cloud Deployment of Big Data Management*

The following image shows the architecture of the Big Data Management deployment on the Azure cloud platform:




The Informatica domain and domain repositories are deployed in the Azure environment.

The Informatica Developer can access the domain from any network.

**Note:** To access the domain from a location outside the VNet where the domain is deployed, you add the domain host name and IP address to the `/etc/hosts` file on the client machine.

# Big Data Management Components

This section contains a description of Big Data Management, its components, and major features.

## *Informatica Domain*

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

### Application Services

**Model Repository Service**

The Model Repository Service manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

**Data Integration Service**

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

**Metadata Access Service**

The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata. The Metadata Access Service contains information about the Service Principal Name (SPN) and keytab information if the Hadoop cluster uses Kerberos authentication.

The Informatica domain can run several other services. For more information about Informatica services, see the *Informatica Application Service Guide*.

### Domain Repositories

Informatica repositories, posted on SQL databases, store metadata about domain objects. Informatica repositories include the following:

**Domain configuration repository**

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

**Model repository**

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings and workflows.

**Monitoring Model repository**

The monitoring Model repository stores statistics for Data Integration Service jobs. You configure the monitoring Model Repository Service in the domain properties.

In addition to these domain repositories, the solution also requires a repository for Hive metadata. This repository is hosted on an SQL database. It stores Hive table metadata to enable Hadoop operations.

For more information about domain repositories, see the *Informatica Application Service Guide*.

## *Informatica clients*

You can use several different clients with Informatica Big Data Management:

**Administrator tool**

The Administrator tool enables you to create and administer services, connections, and other domain objects.

**Developer tool**

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

**Command line interface**

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

## *Functionality*

When you deploy Big Data Management in the Azure cloud environment, you can run mappings on a compute cluster against sources and targets in Azure storage.

This section describes Informatica features that enhance your ability to develop mappings and manage production costs.

### Connectivity

You can use the following adapters to connect to Azure storage resources:

**PowerExchange Adapter for Microsoft Azure Blob Storage**

You can use PowerExchange for Microsoft Azure Blob Storage to connect to Microsoft Azure general purpose storage, known as blob storage.

For more information, see the *Informatica PowerExchange for Microsoft Azure Blob Storage Guide*.

**PowerExchange Adapter for Microsoft Azure Data Lake Store**

Use PowerExchange for Microsoft Azure Data Lake Store to read data from and write data to the Azure Data Lake Store (ADLS). You can collate and organize the details from multiple input sources and use adapter to write data to ADLS.

For more information, see the *Informatica PowerExchange for Microsoft Azure Data Lake Store Guide*.

**PowerExchange Adapter for Microsoft Azure SQL Data Warehouse**

Use PowerExchange for Microsoft Azure SQL Data Warehouse to read data from and write data to Azure SQL Data Warehouse. You can also use the adapter to collate and organize the details from multiple input sources and write the data to Azure SQL Data Warehouse.

For more information, see the *Informatica PowerExchange for Microsoft Azure SQL Data Warehouse Guide*.

### Transformations

Informatica Developer provides a set of transformations that perform specific functions. For example, an Aggregator transformation performs calculations on groups of data.

Transformations in a mapping represent the operations that the Data Integration Service performs on the data. Data passes through transformation ports that you link in a mapping or mapplet.

For more information about transformations that you can use in Big Data mappings, see the *Informatica Developer Transformation Guide*.

### Data Processing Frameworks

You can choose to run mappings on several different run-time engines.

Big Data Management supports the following run-time engines for data processing:

- Spark
- Blaze
- Databricks Spark

On an HDInsight cluster, you can choose the Hadoop option to run mappings on the Blaze engine, or on the Spark engine if the mapping cannot run on the Blaze engine. If you use a Databricks cluster, you must choose the Databricks Spark engine.

### Cluster Workflows (Ephemeral Clusters)

You can use a workflow to create a cluster that runs Mapping and other tasks on a cloud platform cluster.

A cluster workflow contains a Create Cluster task that you configure with information about the cluster to create. The cluster workflow uses other elements that enable communication between the Data Integration Service and the cloud platform, such as a cloud provisioning configuration and a Hadoop connection.

If you want to create an ephemeral cluster, you can include a Delete Cluster task. An ephemeral cluster is a cloud platform cluster that you create and use for running mappings and other tasks, then terminate when tasks are complete to save cloud platform resources.

For more information about creating and deploying cluster workflows, see the following article on the Informatica Network: How to Create Cloud Platform Clusters Using a Workflow in Big Data Management.

## Azure Cloud Platform Elements

The Microsoft Azure cloud platform is composed of the following elements:

### *Container for the Compute Cluster and Other Resources*

A container on the Azure platform provides secure communication between its elements, including a compute cluster.

If you plan to use an HDInsight cluster, the container is a virtual network or VNet. If you plan to use a Databricks cluster, the container is a Databricks workspace.

### Virtual Network (VNet)

A virtual network, or VNet, is a set of virtual machines (VMs) within which you can administer users and groups, security policies, and other policies. The VNet is a logical container for a set of resources. It enables Azure virtual machines (VM) to communicate with each other, and with the internet.

Within the VNet, you can assign resources. You can also segment the VNet into subnets. Each subnet in a VNet is a logical endpoint and is identified by an IP address. For more information, see Azure documentation.

The VNet that you create for the Big Data Management implementation contains the following elements:

- Various storage elements, such as ADLS and blob (or “general”) storage
- Azure HDInsight cluster

- Azure SQL databases, used for Informatica domain repositories
- Azure SQL Data Warehouse
- Network security group

### Databricks Cluster Workspace

If your solution includes a Databricks cluster, you create a workspace in the Databricks service. Associate storage resources and processing nodes with the cluster in this workspace. See the Azure Databricks documentation.

## *Azure Virtual Machines (VMs)*

Virtual machines in the Azure cloud environment act as nodes in the virtual network.

Specify and create Linux VM nodes on the Azure cloud platform to manage and process data. You can choose from several available Linux VM sizes and types that offer a variety of processor speeds.

### Storage Hardware Types for Azure VMs

For each of the following storage types, you can also specify the storage hardware to attach to VMs:

**Hard disk drives (Standard HDD)**

Hard disk drives (standard HDD) are backed by magnetic drives and are preferable for applications where data is accessed infrequently.

Azure standard storage accounts use HDD storage.

**General-purpose solid-state drives (Premium SSD)**

General-purpose solid-state drives (Premium SSD), also known as GP2 volumes, are intended to balance low cost and performance for a variety of transactional workloads.

You can choose between premium and standard SSD. Premium disks are recommended for high IOPS workloads.

Azure premium storage accounts use SSD storage.

### Physical storage

The Big Data Management integration with HDInsight provides native, high-volume connectivity to each of the following storage types:

**Azure Data Lake Storage (ADLS Gen1)**

Azure Data Lake Storage provides massively scalable data storage optimized for Hadoop analytics engines. You can use ADLS to archive structured and unstructured data, and access it via Hive, Spark, or the native Informatica run-time engine.

**General purpose storage v1 and v2**

General purpose storage is available in v.1 (GPv1) and v. 2 (GPv2). Both come in standard and premium versions. The standard storage version uses magnetic media tape storage. Only standard storage supports Hadoop. For more information about general purpose storage, see the Azure documentation.

Use this disk storage for data sources and targets.

### Storage Type Features

The following table describes features of each storage type:

| Storage Type | Description | Accessibility |
|---|---|---|
| Storage (General purpose v1) | - Does not have latest features of Azure storage.<br>- More expensive per-gigabyte pricing model.<br>- Pricing is lower for transactions.<br>- Oldest type of storage account.<br>- Supports blob, files, queues and tables. | Accessible from all Azure storage services |
| Storage v2 (General purpose v2) | - Supports hot, cool, and archive storage.<br>- Supports lowest per-gigabyte pricing model.<br>- When you create a new storage account, General Purpose v2 is the default option.<br>- Recommended for source and target data<br>- Supports blob, Azure files, messages, queues, and un-managed disks (page blobs). | Accessible from all Azure storage services |
| ADLS | - Uses Apache Hadoop.<br>- WebHDFS file system compatible.<br>- No limits on account sizes, file sizes, or the amount of data that can be stored in a data lake.<br>- Individual files can range from kilobyte to petabytes in size.<br>- Performance-tuned for big data analytics.<br>- Can store any data in their native format. | Accessible from all Azure storage services<br>**Note:** You can access ADLS through the Azure API. An HDInsight cluster is not required. |

For more information, see the Azure documentation.

**Note:** Blob storage is also available, but it does not support HDInsight.

### Storage Type Security

The following table describes security characteristics of each storage type:

| Storage Type | Description |
|---|---|
| Storage (General purpose v1) | Uses Resource Manager Role-Based Access Control (RBAC) with storage account keys. |
| Storage v2 (General purpose v2) | Uses Resource Manager Role-Based Access Control (RBAC) with storage account keys. |
| ADLS | - POSIX-compliant fine-grained ACL support.<br>- At-rest encryption.<br>For more information, see the Azure documentation.<br>- Azure Active Directory integration.<br>- Storage account firewalls. |

### Storage Tier Support for Compute Clusters

The following table shows which storage types are supported with HDInsight and Databricks clusters:

| Storage Account Type | Storage Tier | Supported with HDInsight | Supported with Databricks |
|---|---|---|---|
| General purpose storage | Standard | Yes | Yes |
| General purpose storage | Premium | No | Yes |
| Blob storage | Hot / Cool | No | Yes |
| ADLS | Gen 1 | Yes | Yes |

For more information about storage types to use with HDInsight, see the Azure documentation here. For more information about storage that you can use with Databricks, see the Azure documentation here.

## *Compute Cluster Nodes*

Clusters on the Azure platform are organized with a head or driver node that helps manage the cluster, and worker nodes that process jobs.

### HDInsight Cluster Nodes

HDInsight clusters are made up of the following types of nodes, each of which corresponds to a VM:

**Head node**

The head node of a cluster controls distribution of processing tasks to other nodes in the cluster. The head node runs Hadoop services, including HDFS, YARN, Hive metastore, application timeline server (ATS), and others.

**Worker nodes**

The worker nodes in a cluster perform processing tasks. You can specify any number of worker nodes.

You can manually scale the number of worker nodes up or down. HDInsight does not support auto-scaling.

**Gateway nodes**

In addition to the head and worker nodes, each cluster includes two gateway nodes that run management and security tasks. Users do not have access to gateway nodes.

For more information about HDInsight cluster nodes, see the Azure HDInsight documentation.

### Databricks Cluster Nodes

Databricks clusters have a driver node instead of a head node. Worker nodes make up the remainder of the cluster.

For more information about Databricks cluster nodes, see the Azure Databricks documentation.

## Node Types for Cluster Workflows

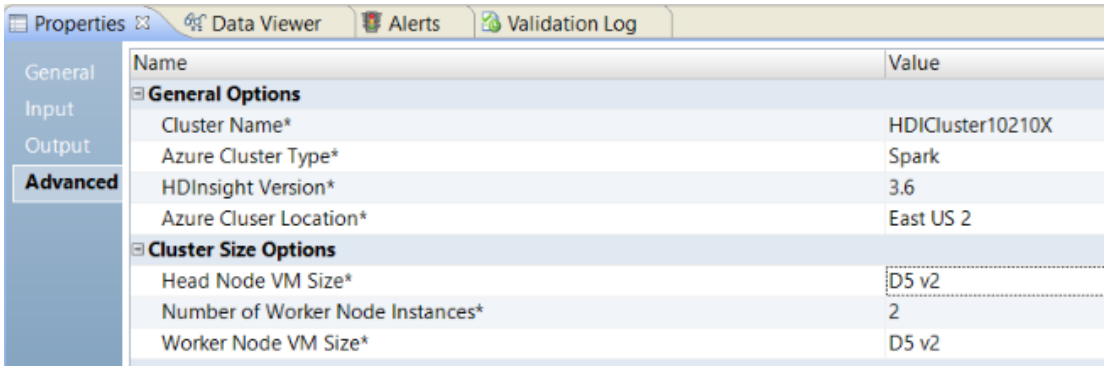
Cluster workflows enable you to automate the creation of cluster and run specified mappings. The cluster workflow can include a task to delete the cluster when processing is complete.

For more information about cluster workflows, see the *Big Data Management User Guide*. The following node types support cluster workflow operations on HDInsight:

- A series nodes:
  - A3
  - A4
  - A7
- DS_v2 series nodes:
  - D5_v2
  - D12_v2
  - D13_v2
  - D14_v2

  **Note:** DS_v2 series nodes are General Purpose type.

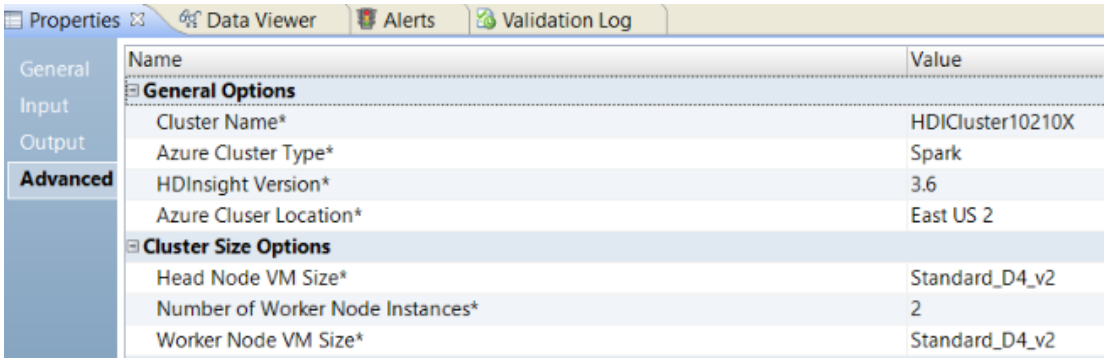
The following image shows the Cluster Size Options properties in the Advanced properties tab of the create cluster task:




You can also specify other cluster node types when you configure the Create Cluster task. Manually enter a valid node type. For example, type `Standard_D4_v2`.

For a list of valid node types, and more information about the specifications of available cluster node types, see the Azure documentation.

The following image shows an example where the user has manually typed an alternate node size for the Head Node VM Size and Worker Node VM Size properties:



## *Azure SQL Cloud Database*

The Azure SQL Database (DB) is a general-purpose relational database managed service in Microsoft Azure that supports structures such as relational data, JSON, spatial, and XML.

Azure SQL DB offers managed single SQL databases and managed SQL databases in an elastic pool. You can scale database transaction units (DTU) and storage sizes, and scale up or down to different database types.

### SQL Database Purchasing Models

Choose between the following purchasing models for SQL databases:

**Database Transaction Unit-based model**

The Database Transaction Unit (DTU) is based on a blended measure of CPU, memory, reads, and writes. The DTU-based performance levels represent preconfigured bundles of resources to drive different levels of application performance.

**vCore-based model**

Customers who need more insight into the underlying resources or need to scale them independently to achieve optimal performance should consider a vCore-based model. A vCore or virtual core represents the logical CPU. You can choose among hardware configurations for vCores.

### SQL Database Types

Choose from among Basic, Standard, or Premium Azure SQL database types.

**Basic**

The Basic service tier offers good performance in a small database to which you make few concurrent requests.

**Standard**

The Standard service tier offers databases that support multiple concurrent requests. You can size the Standard service tier database application based on predictable performance, minute over minute.

**Premium**

The Premium service tier supports business-critical databases. You can size the Premium service database application based on the peak load for that database. The plan removes cases in which performance variance can cause small queries to take longer than expected in latency-sensitive operations.

### Repository Database Storage

Informatica repositories, such as the domain repository, Model repository, and monitoring Model repository, require their own storage. You can use an on-premises SQL database or the recommended Azure SQL database for domain repositories.

Azure SQL databases use dedicated storage, which is neither ADLS nor blob storage. When you choose the type and size of database instance, Azure dedicates this storage to the domain repositories.

# How to Deploy and Integrate Big Data Management on the Azure Cloud Platform

This section contains a summary of the tasks you perform to set up Big Data Management to run mappings and workflows on the Azure cloud platform.

### Manual Deployments

When you perform a manual cloud or hybrid deployment of Big Data Management on Azure, you perform the following steps:

Step 1. Verify prerequisites.

Step 2. Prepare the Azure environment.

Step 3. Install and integrate the Informatica domain in the Azure environment.

This section provides a summary of these tasks. For details on how to perform these tasks see the Big Data Management 10.2.2 documentation set, available for download from the Informatica Documentation Portal.

**Note:** As you create and configure Azure resources and the Informatica domain, keep in mind that you must create the following elements in the same Azure location:

- Azure SQL database
- Azure VM
- Compute cluster
- Informatica domain

### Marketplace Deployments

In contrast with manual deployments, a marketplace deployment is largely automated.

This article does not describe how to perform marketplace deployments. To see how to implement a marketplace deployment of Big Data Management on the Azure cloud platform, see the article "Deploying Big Data Management 10.2.2 on Microsoft Azure through the Azure Marketplace".

## *Step 1. Verify Prerequisites*

Verify the following prerequisites:

- You have purchased a license for Big Data Management and have a download link to the Informatica installer. The license file has a name like `BDMLicense.key`. During configuration, you browse to this file on your local system. It is not necessary to upload it to the Azure environment.
- You have an Azure subscription and its account authentication information.
  This article assumes you have experience administering resources on the Azure cloud platform.

## *Step 2. Prepare the Azure Environment*

Create resources on the Azure cloud platform to contain and provide the infrastructure for Big Data Management operations.

Read Azure documentation for detailed instructions for these tasks.

### Step 2.1. Create a Resource Group

A resource group is a container that holds related resources for an Azure solution. The resource group is a holder for metadata about the resources, such as storage accounts, databases, and other objects. You can use tags on objects to denote them as members of the resource group.

Create a resource group for the Big Data Management implementation to contain metadata about the following resources for the solution to access:

- Network interface
- Network security group
- Virtual network (VNet)
- Public IP address
  Choose to create a static IP address, not a dynamic one.
- Storage resources, including ADLS, OS-level disks, and general purpose storage
- Database resources
- Virtual machines
- HDInsight clusters

### Step 2.2. Create a Container

Depending on the type of cluster you plan to use, create a container for the cluster and other resources.

If you plan to use an HDInsight cluster, create a VNet to contain the Big Data Management implementation. For more information, see the Azure documentation on virtual networks.

If you plan to use a Databricks cluster, create a workspace for Databricks. For more information, see the Azure Databricks documentation.

### Step 2.3. Create the SQL Databases for Repositories

Implementing Big Data Management on the Azure cloud platform requires an SQL server database for the domain configuration repository, the Model repository, the monitoring Model repository, and the Hive metastore.

Set up separate databases for each repository.

Choose between the Azure SQL Database on the Azure cloud platform or an on-premises RDBMS system. Big Data Management supports an on-premises SQL Server database, but Informatica recommends using the Azure SQL database for best performance.

### Step 2.4. Create VMs

Create virtual machines (VMs) in the VNet.

#### VMs and Storage

Create the VMs as members of the resource group that you created.

Attach physical storage to the VMs to create space for the Informatica installer and the Big Data Management deployment. The installer and installed domain require 100 GB of storage.

Specify the type of storage to store and retrieve data. Choose between HDD and SSD storage, depending on your use case. See "Azure Virtual Machines (VMs)" on page 8 for information about the types of physical storage available.

Informatica recommends the following configuration for physical storage:

- Managed disks to enable data redundancy and fault tolerance
- Standard HDD as the VM disk type for OS disks
- Enable host caching for both OS and data disks

### Regions and Availability Zones

Azure supports a set of regions which correspond to geographies or other specifications. View the list of supported regions in Azure documentation at the following URL:
https://azure.microsoft.com/en-us/global-infrastructure/regions/.

Availability zones are physically separate locations within a region. Each availability zone is made up of one or more physical data centers. You can locate your big data infrastructure within a single availability zone to reduce network latency. Read more about availability zones at the following URL:
https://azure.microsoft.com/en-us/global-infrastructure/availability-zones/.

## Step 2.5. Copy the Informatica Installer Binaries and Run the Installer

Download the Informatica installer binaries, and then copy them to Azure storage and run the Informatica installer.

Install the Informatica domain on **standard HDD data disks**. You can use premium SSD disks if your use case requires high performance.

The following table gives some details about performance of HDD and SSD disks:

| Feature | Standard HDD | Premium SSD |
|---|---|---|
| IOPS limit | 500 | 5000 |
| Throughput limit (MB/sec) | 60 | 200 |

For information about installing and configuring the Informatica domain, see the following Informatica documentation:

- *Big Data Suite Installation Guide*
- *Big Data Management Administrator Guide*
- *Informatica Application Service Guide*

You can access Informatica documentation on the Informatica Network.

## Step 2.6. Create or Identify Storage Resources

Create or identify existing Azure storage resources to use with the Big Data Management implementation.

Create or identify SQL databases for the following data:

- The following Informatica repositories:
  - Domain repository
  - Model repository

- Monitoring Model repository

- Hive metadata repository
  The Hive metadata repository stores definitions for Hive table schemas. Use an Azure SQL database for the metadata repository, and attach the database to the HDInsight cluster during cluster creation.

You can create or use existing ADLS, WASB, or general storage to serve as the repository for source and target data.

After you create storage resources, upload data to Azure storage, and use Hive queries to create Hive tables. See Azure documentation at the following URL:
https://docs.microsoft.com/en-us/azure/machine-learning/team-data-science-process/move-hive-tables.

### Step 2.7. Provision a Compute Cluster

Provision a Databricks or HDInsight compute cluster in the resource group that you created for the Big Data Management implementation.

When you provision an HDInsight cluster, attach a Hive metastore database. The Hive metastore is a semantic repository that stores metadata for Hive tables, enabling Hadoop operations on the cluster.

**Note:** The cluster must be in the same region as the storage that it accesses. You provisioned this storage in step 2.6.

## *Step 3. Install and Configure the Informatica Domain*

Install and configure the Informatica domain.

You can deploy Big Data Management on the Azure cloud platform or on-premises. When you create the domain in the Azure cloud platform, it must be in the same location as the Azure resources deployed for the project.

For more information on deploying and configuring Big Data Management, see the *Big Data Management Integration Guide*.

# Sizing and Tuning Big Data Management Deployments

Sizing and tuning recommendations vary based on the deployment type. Based on certain deployment factors in the domain and Hadoop environments, Informatica categorizes Big Data Management into the following types:

**Sandbox deployment**

Used for proof of concept engagements, or as a sandbox with minimal users.

**Basic deployment**

Used for low-volume processing with low levels of concurrency.

**Standard deployment**

Used for high-volume processing with low levels of currency.

**Advanced deployment**

Used for high-volume processing with high levels of concurrency.

## *Big Data Management Deployment Type: Deployment Criteria*

The following criteria determine the Big Data Management deployment type:

**Number of active users**

Number of users working on the Model repository at design time, using the Analyst tool, or running Big Data Management jobs in the native or Hadoop run-time environment at any given point of time.

**Number of concurrent pushdown mappings**

Total number of mappings running on the Blaze, Spark, or Hive engines that are concurrently submitted to the Data Integration Service.

**Number of objects in the Model repository**

Total number of design-time and run-time objects in the Model repository. For example, data objects, mappings, workflows, and applications.

**Number of deployed applications**

Total number of applications deployed across all the Data Integration Services in the Informatica domain.

**Number of objects per application**

Total number of objects of all types that are deployed as part of a single application.

**Total operational data volume**

Total volume of data being processed in the Hadoop environment at any given point of time.

**Total number of data nodes**

Total number of data nodes in the Hadoop cluster.

## *Big Data Management Deployment Type: Deployment Type Comparison*

The following tables compare Big Data Management deployment types based on the standard values for each deployment factor:

### Domain environment

The following table contains guidelines for deployment factors for the domain environment:

| Deployment Factor | Sandbox Deployment | Basic Deployment | Standard Deployment | Advanced Deployment |
|---|---|---|---|---|
| Number of active users | 1 | 5 | 10 | 50 |
| Number of concurrent pushdown mappings | < 10 | 20 -500 | 500 - 1000 | > 1000 |
| Number of objects in the Model repository | < 1000 | < 5000 | < 20,000 | 20,000 + |
| Number of deployed applications | < 10 | < 25 | < 100 | < 500 |
| Number of objects per application | < 10 | 10 - 50 | 50 -100 | 50 -100 |
| Total operational data volume (for batch processing use cases) | 10 GB | 100 GB | 500 GB | 1 TB + |

### Hadoop Environment

The following table contains guidelines for deployment factors for the Hadoop environment:

| Deployment Factor | Sandbox Deployment | Basic Deployment | Standard Deployment | Advanced Deployment |
|---|---|---|---|---|
| Total number of worker nodes | 3 | 5 - 10 | 10 - 50 | 50 + |
| yarn.nodemanager.resource.cpu-vcores | 12 | 24 | 24 | 36 |
| yarn.nodemanager.resource.memory-mb | 12288 MB | 24576 MB | 49152 MB | 98304 MB |

Based on the deployment type that you use to categorize Big Data Management, you can use `infacmd autotune autotune` to automatically tune certain properties in your environment.

For more information, see the "Tuning for Big Data Processing" chapter in the *Informatica Big Data Management Administrator Guide*.

## Sizing: Hadoop Cluster Hardware Recommendations

The following table lists the minimum and optimal hardware requirements for the Hadoop cluster:

| Hardware | Sandbox Deployment | Basic or Standard Deployment | Advanced Deployment |
|---|---|---|---|
| Logical or virtual CPU cores | 16 | 24 - 32 | 48 |
| Total system memory | 16 GB | 64 GB | 128 GB |
| Number of nodes | 2+ | 4 - 10+ | 12+ |

## Sizing: Informatica Domain

This section describes several considerations and recommendations for configuring VMs dedicated to the Informatica domain.

### Deployment Types

The following table contains guidelines for choosing the deployment type for the domain:

| Deployment Type | Use Case | Total Virtual Cores | Ram Per Node | Disk Space Per Node[1] |
|---|---|---|---|---|
| Sandbox | Proof of concept, or a sandbox environment with minimal users | 16 | 32 GB | 50 GB |
| Basic | Low-volume processing, low concurrency | 24 | 32 GB | 100 GB + |
| Standard | High-volume processing, low concurrency | 48 | 64 GB | 100 GB + |

| Deployment Type | Use Case | Total Virtual Cores | Ram Per Node | Disk Space Per Node[1] |
|---|---|---|---|---|
| Advanced | High-volume processing, high concurrency | 96 | 64 GB | 100 GB + |

[1] Disk space requirement is for Informatica services. Additional disk capacity is required to process data in the native run-time environment.

## *Sizing: SQL Database VM Instances*

This section describes several considerations and recommendations for configuring VMs dedicated to SQL databases.

SQL databases are required to contain the domain repository, the Model repository, the monitoring Model repository, and the Hive metadata repository.

Azure SQL databases are categorized as Basic, Standard, or Premium. These categories are based on database transaction units (DTU). Higher DTU means higher performance, as well as higher cost. You should balance performance needs with your budget.

The following table shows a range of Standard instances:

| Instance Level | DTU |
|---|---|
| S1 | 20 |
| S2 | 50 |
| S3 | 100 |
| S4 | 200 |
| S5 | 400 |

Standard DB instances, which are available in a range of instance sizes, will suit most of your needs. For I/O-intensive workloads, you can scale up to a Premium DB type.

The following table lists the minimum DTU requirements for the VM on which each repository database resides:

| Repository | Sandbox | Basic | Standard | Advanced |
|---|---|---|---|---|
| Domain | 50 | 50 | 100 | 200 |
| Model | 100 | 200 | 400 | 800 |
| Monitoring | 100 | 200 | 400 | 800 |
| Hive metadata | 50 | 50 | 100 | 200 |

# Best Practices

To achieve the best performance for Big Data Management on the Azure cloud, consider the following best practices and recommendations:

- Create the following resources in the same geographic location and VNet:
  - Azure SQL databases for the domain, monitoring, and Model repositories
  - Azure VM for the Informatica domain
  - Azure Storage (ADLS or GPv2)
  - HDInsight cluster
  - Azure Windows VM with Developer tool installation
- Use the private IP address when configuring the connection to the following host instances in the Azure cloud:
  - Database hosts
  - Instances that host Informatica domain services
- Choose between ADLS or General-Purpose Storage (GPv2) for persistent data storage, depending on your use case. For example, ADLS is more commonly used for a data analytics use case.
- With data residing in ADLS or GPv2, you can terminate the HDInsight cluster with a Delete Cluster task after the job is completed, providing significant cost savings.
- To replicate data in Azure Storage in different locations, use cross-regional replication with RA-GRS. RA-GRS replicates your data to another data center in a secondary region and also provides you with the option to read from the secondary region. See the Azure documentation.
- Spark shuffle service is enabled by default if you select Spark as the cluster type during the HDInsight cluster configuration process. Chose Spark version 2.3.0 (HDI 3.6).

## *Configure Preemption for Concurrent Jobs*

Configure the Databricks cluster to improve concurrency of jobs.

When you submit a job to Databricks, it allocates resources to run the job. If it does not have enough resources, it puts the job in a queue. Pending jobs fail if resources do not become available before the timeout of 30 minutes.

You can configure preemption on the cluster to control the amount of resources that Databricks allocates to each job, thereby allowing more jobs to run concurrently. You can also configure the timeout for the queue and the interval at which the Databricks Spark engine checks for available resources.

Configure the following environment variables for the Databricks Spark engine:

**spark.databricks.preemption.enabled**

Enables the Spark scheduler for preemption. Default is false.

Set to: true

**spark.databricks.preemption.threshold**

A percentage of resources that are allocated to each submitted job. The job runs with the allocated resources until completion. Default is 0.5, or 50 percent.

Set to a value lower than default, such as 0.1.

**spark.databricks.preemption.timeout**

The number of seconds that a job remains in the queue before failing. Default is 30.

Set to: 1,800.

**Note:** If you set a value higher than 1,800, Databricks ignores the value and uses the maximum timeout of 1,800.

**spark.databricks.preemption.interval**

The number of seconds to check for available resources to assign to a job in the queue. Default is 5.

Set to a value lower than the timeout.

Changes take effect after you restart the cluster.

**Important:** Informatica integrates with Databricks, supporting standard concurrency clusters. Standard concurrency clusters have a maximum queue time of 30 minutes, and jobs fail when the timeout is reached. The maximum queue time cannot be extended. Setting the preemption threshold allows more jobs to run concurrently, but with a lower percentage of allocated resources, the jobs can take longer to run. Also, configuring the environment for preemption does not ensure that all jobs will run. In addition to configuring preemption, you might choose to run cluster workflows to create ephemeral clusters that create the cluster, run the job, and then delete the cluster. For more information about Databricks concurrency, contact Azure Databricks.

# Case Studies: Azure SQL Database

The following sections contain findings resulting from Azure SQL DB resource usage tests on a Big Data Management deployment using an HDInsight cluster.

The tests measure performance using a unit of measurement, the **Database Transaction Unit** (DTU). The DTU measurement combines measurements of CPU, I/O and memory (RAM) capabilities based on a benchmark OLTP workload. It is a measurement of database throughput that accounts for CPU capacity, memory capacity, and read-write rates. DTU-based performance levels represent preconfigured bundles of resources to drive different levels of application performance.

Except where noted, the tests were conducted on a Standard S6 DB instance with 400 DTU capacity.

## *CRUD Operations Test 1: Single user performing operations*

The test measured the performance of save, browse, open, update, delete and save operations by a single user running a varying number of mapping objects.

The test resulted in the following findings:

- CRUD operations on a standard-sized mapping with 10 transformations require 24-30 DTU, on average.
- Save, update and delete are the most DTU-intensive operations.
- A single user conducting CRUD operations on 30-40 objects approaches the instance DTU limit of 400 DTU.

## *CRUD Operations Test 2: Multiple users performing concurrent operations*

Tests simulated multiple users performing concurrent browse, update, and save operations. For each user, the test measured the performance of one mapping with 10 transformations.

The test resulted in the following findings:

- In testing the Save operation, increasing the number of concurrent users from 10 to 50 resulted in an increase of DTU usage to a peak of 85% of the 400 DTU capacity.
- The higher the number of concurrent users performing save, update and delete operations, the more DTU-intensive the workload is.

## Repository creation and deletion test

The test measured performance as a user created and deleted a Model repository. The test was repeated with encryption enabled and disabled.

The test resulted in the following findings:

- Creating a Model repository was 47% slower with encryption enabled. However, since Model repository creation is a one-time operation, you can enable encryption without a performance penalty during run-time.
- 26% (~104 DTY) of available DTU is used during repository creation.
- 23% (~92 DTU) of available DTU is used during repository deletion.

## Client Operations Test: User Operations from an On-Premises Developer Tool Client

We tested various actions from an on-premises deployment of the Developer tool connecting to a Model repository deployed in the Azure environment.

The test resulted in the following findings:

- For a small to medium mapping with 5-10 transformations, import/export, rename, delete, copy-paste, and open operations take 1-3 seconds. For large mappings, such as those with 100+ transformations, the same operations take 7-21 seconds.
- Large mappings take a disproportionately long time to open in the Developer tool.

**Tip:** You can reduce the time consumed by Developer tool operations by staging the developer tool on a dedicated Windows VM in the Azure environment. Using a Developer tool in the same VNet as the solution also removes the need for adding an entry in the client machine hosts file.

## Application deployment test

The test measured the deployment speed of applications of varying sizes from an on-premises Developer tool to the Data Integration Service posted on the Azure environment.

The test resulted in the following findings:

- Application deployment is resource intensive. Application deployment occupied 14% of the DTU capacity (~60 DTU) when deploying an application with 40 small mappings containing 1 to 5 transformations.
- The usage curve applies equally to a single application with many mappings, or many applications with a smaller number of mappings each.

## Domain Repository Upgrade and Restore Test

The test measured performance of the following database operations: restore and upgrade of repository schema and contents.

The test was conducted against an Azure SQL database on an S7 instance with 800 DTU capacity. The repository backup file size was 1.4 GB.

The test resulted in the following findings:

- The repository restore operation consumed 800 DTU (100% of available DTU).
- The repository upgrade operation consumed 576 DTU (72% of available DTU).

  **Tip:** Before upgrade, increase the DTU capacity to two or three times standard capacity. After the upgrade, decrease the capacity to its previous size.

# Case Studies: HDInsight Cluster

This section describes results from tests that Informatica conducted to measure the performance of various TPC-DS queries on an HDInsight cluster.

Informatica retrieved TPC-DS queries from the tpc.org site and converted them into big data mappings.

The following table describes HDInsight worker and head nodes that Informatica accessed to perform query benchmark tests:

| DS_v2 Series Node Type | vCPU | Memory (GB) | Worker Node Count |
|---|---|---|---|
| Standard_D4_v2 | 8 | 28 | 8 |
| Standard_D5_v2 | 16 | 56 | 8 |

Tests were executed on a system with the following specifications:

- HDInsight v. 3.6 cluster, standard deployment type

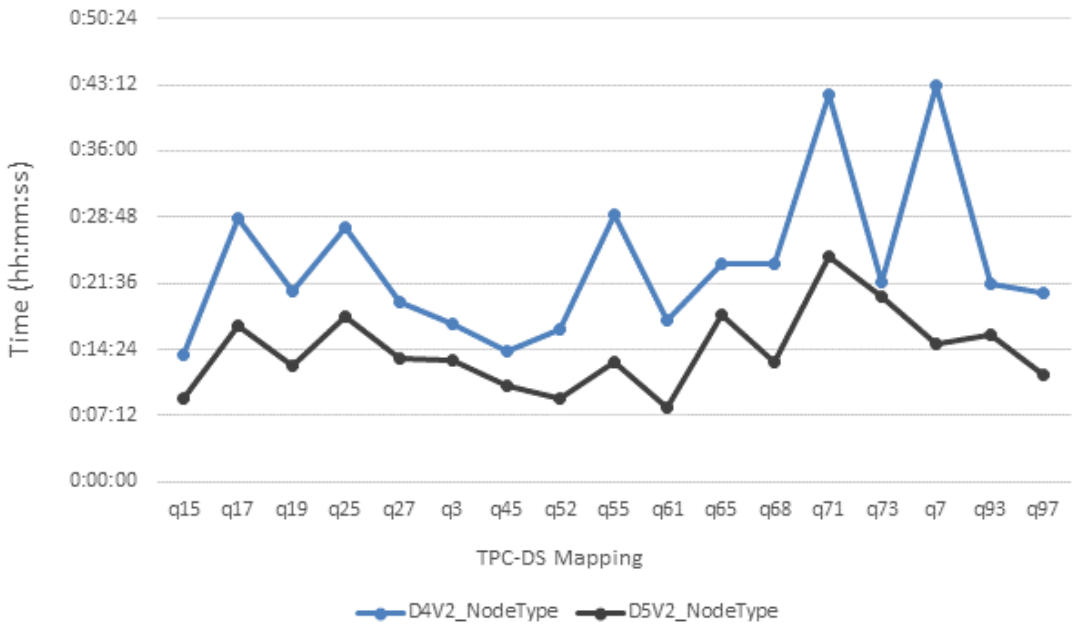
  **Note:** For deployment type descriptions, see "Sizing: Hadoop Cluster Hardware Recommendations " on page 18.
- Spark version 2.2.1

## *Query Execution on Two Different HDInsight Node Types*

The test measured the performance of several TPC-DS queries against sources on two different HDInsight node types accessing Azure Data Lake storage.

The queries were executed using the Spark run-time engine against 1 TB source data. Mapping sources and targets were Hive tables with data residing on ADLS storage.

The following graph shows query performance on VM node types D4v2 and D5v2:



The test resulted in the following findings:

- Mappings executed almost 2X faster on the D5v2 HDInsight node types.

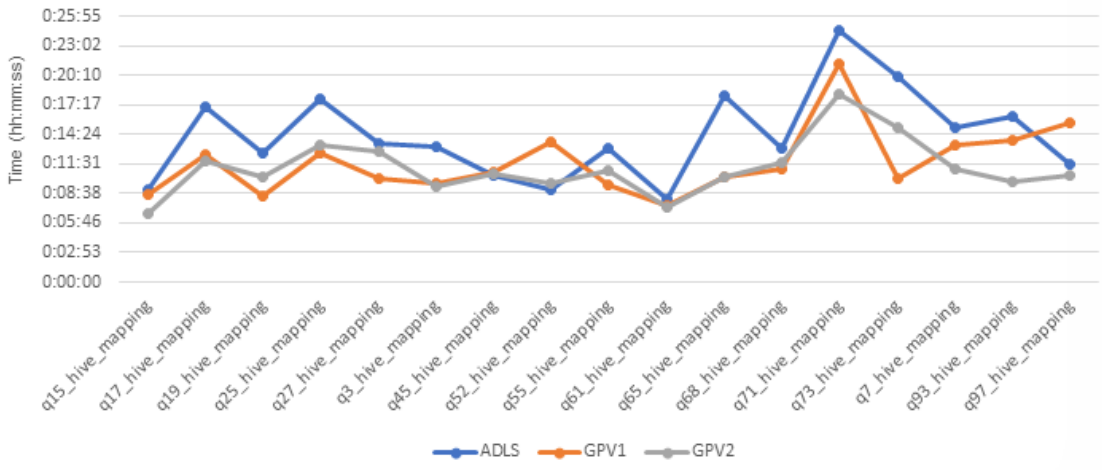
## *Performance of Queries Against Different Storage Types*

Tests measured the performance of queries against the same data set on three different storage types: ADLS Gen1, General purpose v1 (GPv1), and General purpose v2 (GPv2). In each test, the sources and targets were text format files.

### Test 1: TPC-DS Queries on ADLS, GPv1, and GPv2

The test was performed on a D5v2 HDInsight node type using the Spark run-time engine to run the queries against 1 TB source data. Mapping sources and targets were Hive tables with data residing on ADLS, GPv1, and GPv2 storage.

The following graph shows query performance against each storage type:




The test resulted in the following findings:

- Most queries run against data on GPv2 storage were faster.

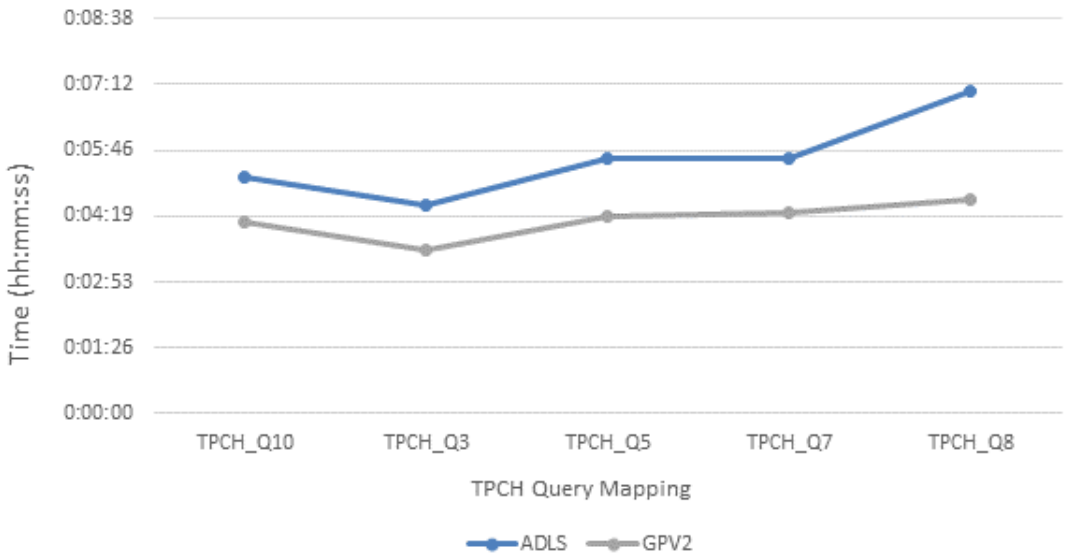
### Test 2: TPCH Queries on ADLS and GPv2

Informatica retrieved TPCH queries from the tpc.org site and converted them to big data mappings.

Tests were performed on a D4v2 HDInsight node type using the Spark runtime engine to run the queries against a TPCH scale factor 100 GB data source.

**Note:** Due to the extended time it takes to run a TPCH query, we chose a smaller 100 GB data set for this test.

The following graphs contrast the performance of queries against ADLS and GPv2 sources:




The tests resulted in the following findings:

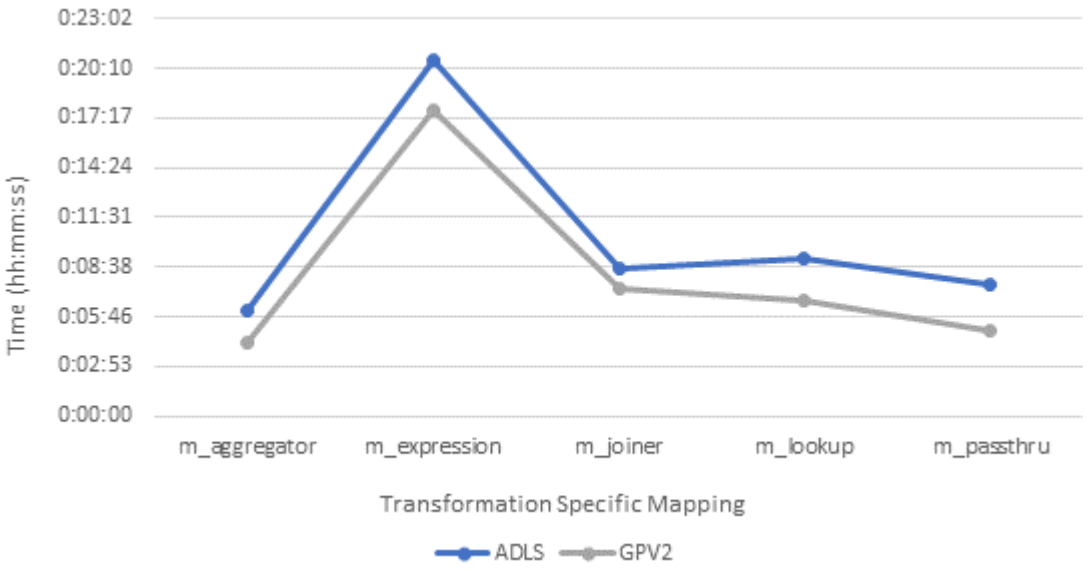
- Accessing data from a GPv2 source is generally 15-30% faster than accessing data from an ADLS source.

### Test 3: Transformation-Specific Mappings

The following table shows the sources used for the mappings used in transformation-specific mapping tests:

| Mappings | Data Source Tables | Data Source(s) Total Size |
|---|---|---|
| - m_aggregator<br>- m_expression<br>- m_passthru | - LineItem | 75 GB |
| - m_joiner<br>- m_lookup | - LineItem<br>- Orders | 92 GB |

The following graph shows the results of the transformation-specific mapping tests:




## Case Studies: Databricks Cluster

The following sections contain findings from tests on a Big Data Management deployment using a Databricks cluster.

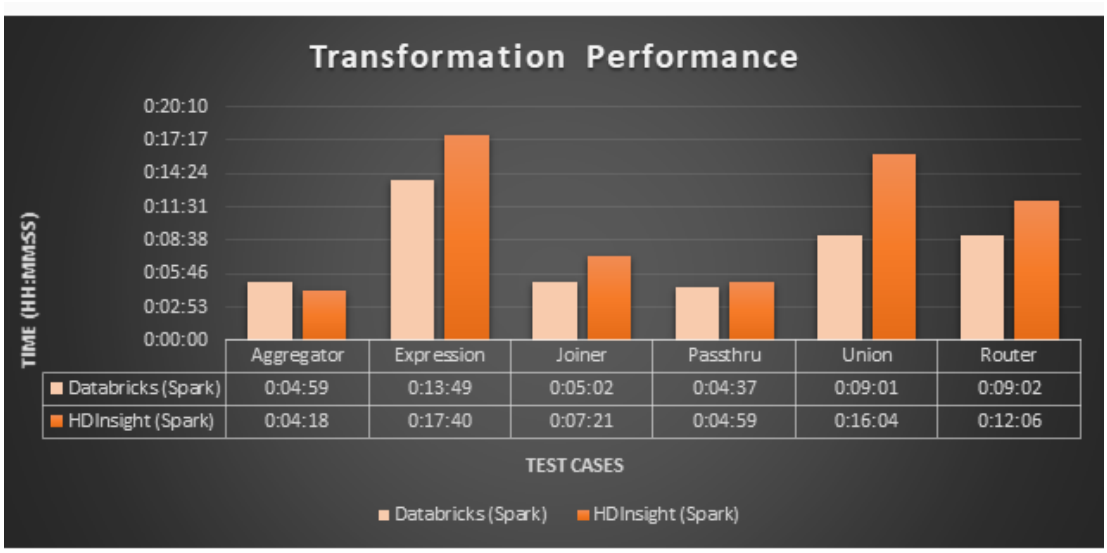
Sources and targets were stored on Azure blob storage. Except where noted, the tests were conducted on a Standard S8_DS4_v2 instance with the following configuration:

- 28 GB RAM
- 8 cores
- 1.5 DBU

## Transformation Performance Comparison

The test compared the performance of several different transformations on HDInsight and Databricks clusters. The test used a TPCH SF100 data volume.

The following image shows test results:




| | Aggregator | Expression | Joiner | Passthru | Union | Router |
|---|---|---|---|---|---|---|
| Databricks (Spark) | 0:04:59 | 0:13:49 | 0:05:02 | 0:04:37 | 0:09:01 | 0:09:02 |
| HDInsight (Spark) | 0:04:18 | 0:17:40 | 0:07:21 | 0:04:59 | 0:16:04 | 0:12:06 |

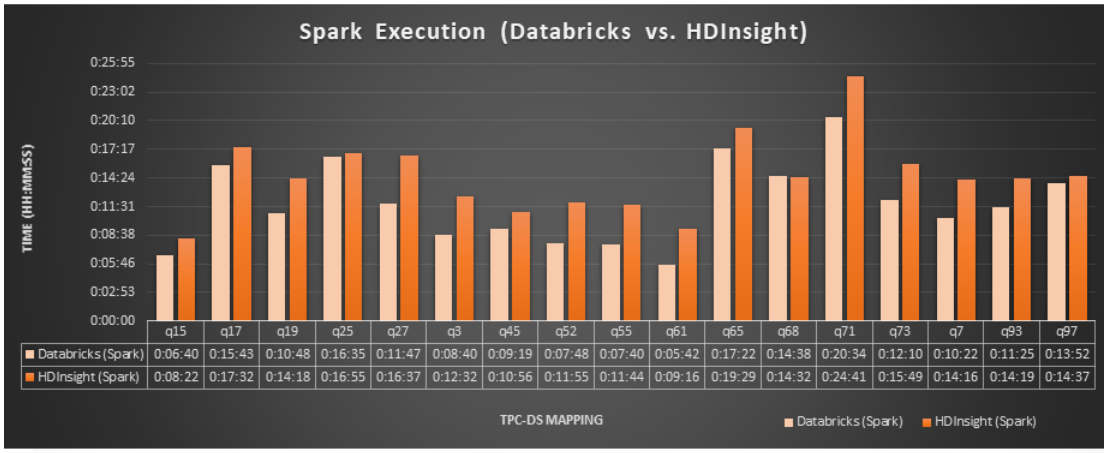
The test resulted in the following findings:

- On average, mappings being processed on a Databricks cluster run 19% faster than on an HDInsight cluster.

## TPC-DS Query Mapping Comparison

The test compared the execution speed of TPC-DS query mappings on HDInsight and Databricks clusters.

The following image shows test results:




| | q15 | q17 | q19 | q25 | q27 | q3 | q45 | q52 | q55 | q61 | q65 | q68 | q71 | q73 | q7 | q93 | q97 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Databricks (Spark) | 0:06:40 | 0:15:43 | 0:10:48 | 0:16:35 | 0:11:47 | 0:08:40 | 0:09:19 | 0:07:48 | 0:07:40 | 0:05:42 | 0:17:22 | 0:14:38 | 0:20:34 | 0:12:10 | 0:10:22 | 0:11:25 | 0:13:52 |
| HDInsight (Spark) | 0:08:22 | 0:17:32 | 0:14:18 | 0:16:55 | 0:16:37 | 0:12:32 | 0:10:56 | 0:11:55 | 0:11:44 | 0:09:16 | 0:19:29 | 0:14:32 | 0:24:41 | 0:15:49 | 0:14:16 | 0:14:19 | 0:14:37 |

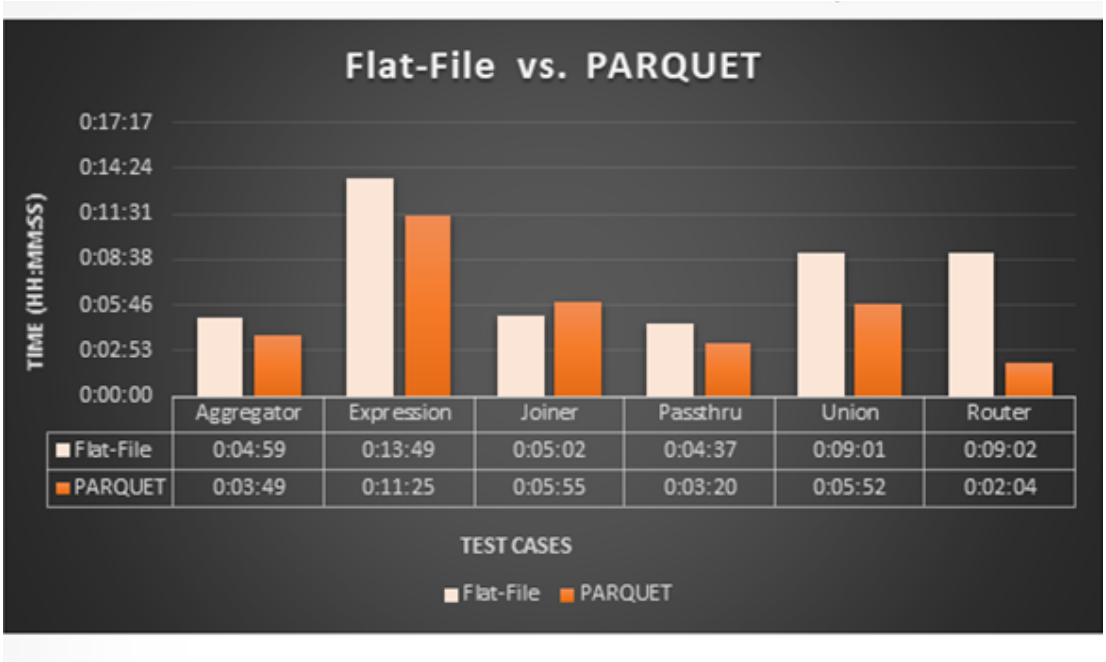
The test resulted in the following findings:

- On average, TPC-DS query mappings being processed on a Databricks cluster run 20% faster than on an HDInsight cluster.

## Comparison Between Flat-File and Parquet Sources and Targets

The test compared the execution speed of mappings that access parquet sources and targets with mappings that access flat-file sources and targets.

The following image shows test results:




The following list shows the disk usage optimization flags that affected the tests:

- spark.databricks.io.cache.enabled
- spark.databricks.io.cache.maxDiskUsage
- spark.databricks.io.cache.maxMetaDataCache
- spark.databricks.io.cache.compression.enabled

**Note:** The spark.databricks.io.cache.enabled flag applies only to parquet files and does not affect flat file performance.

The test resulted in the following findings:

- Databricks processes mappings 27% faster with default settings when the mappings access parquet sources and targets.
- Optimization techniques for a Databricks cluster are applicable only to mappings that access parquet sources and targets.

## When to Use Auto-Scaling with Databricks Clusters

Enable auto-scaling when you anticipate mixed loads on the compute cluster. Auto-scaling is recommended for long-running tasks or multiple tasks that run for long periods.

When you use auto-scaling, Databricks automatically determines when to upsize or to downsize.

Using auto-scaling allows you to control usage costs. The cluster brings up additional nodes only when they are needed for resource-intensive tasks.

Note the following behavior patterns:

- Once auto-scaling starts, Databricks takes 2–4 minutes to commission additional nodes.
- Databricks automatically downsizes when jobs are completed; this process takes 2–3 minutes

## Authors

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